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Blood compatibility of heparin-inspired, lactose containing, polyureas depends on the chemistry of the polymer backbone†

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The influence of polymer chemical structure and degree of sulfation of polyurea glycopolymers on blood compatibility is reported. Polymers were prepared using lactose-containing diamines with either toluene 2,4-diisocyanate (TDI), isophorone diisocyanate (IPDI), methylene bis(4-cyclohexyl isocyanate) (HMDI), or hexamethylene diisocyanate (HDI) and subsequntly sulfated with a sulfur trioxide reagent. The influence of degree of sulfation was investigated using a polymer. Polyurea glycopolymers with sulfur contents ranging from ~3% to 15% were prepared from HMDI polymerized with the lactose-containing diamine. The obtained glycopolymers have been characterized using gel permeation chromatography, nuclear magnetic resonance spectroscopy, Fourier transform infrared spectroscopy, and elemental analysis. The blood compatibility of the polymers was determined by measuring the activated partial thromboplastin time, thrombin time, and prothrombin time. Poly(HMDI-sL) exhibited higher prolongation of the activated partial thromboplastin time than other diisocyanate-containing sulfated glycopolymers, and higher degrees of sulfation result in prolonged clotting times. The polymers ability bind thrombin in the presence and absence of antithrombin III was measured using a coloremtric thrombin inhibition assay and the results followed the trend observed in the activated partial thromboplastin time assay.

Introduction

Biomaterials interface with biological systems to evaluate, treat, augment, or replace a body's tissue, organ, or function.[1] Polyurethane-ureas are widely used in biomaterials, and can be found in vascular prostheses, catheters, blood pump, heart valves, blood filters, blood tubing, and artificial hearts.[2–4] These applications of polyurethane-urea materials are enabled by the diverse chemical structures available, the monomer structure influencing the shape, rigidity, and hydrophobicity of resulting polymers. Therefore, different structures produce different physical properties or biocompatibility in resulting polymers. In our work we are focused on polymer blood compatibility. A non-native biomaterial may initiate a cascade of events upon contact with blood involving protein adsorption, cell–biomaterial surface interaction, platelet adhesion and activation, and clot formation.[5–7] Many polyurethane-ureas exhibit poor biocompatibility due to their hydrophobicity.[8] Common strategies for improving biocompatibilities of polyurethane-ureas are incorporating hydrophilic functional groups or known blood-compatible materials, including phosphates,[9–11] sulfonates,[12] peptides,[13,14] poly(ethylene glycol) (PEG),[15–18] or dermatan sulfate.[19]

Heparin has been used as an anticoagulant drug for over 80 years and proposed for use in blood contacting biomaterials.[20] Heparin is a linear, highly sulfated, and negatively charged polysaccharide. However, there are several concerns with using heparin.[21] These concerns include heparin's diverse structure, which results in batch dependent dose–response relationships; risk of pathogen contamination, such as bovine spongiform encephalopathy; potential contamination of therapeutic heparins (*e.g.* the heparin contamination seen in 2008);[22] and possible heparin-induced thrombocytopenia. One solution for mitigating these disadvantages is to synthesize heparin-mimicking polymers, designed to impart the biological functions of heparin while alleviating the inherent safety concerns of a mammalian sourced biomacromolecule.[23–37] Our group has published syntheses of heparin-inspired polymers using step-growth polymerization and chain-growth polymerization.[38–40] Sulfated polyurea glycopolymers with glucose, mannose, lactose or glucosamine groups were syn-

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† Electronic supplementary information (ESI) available: Detailed synthesis of all polymers and associated spectra, DSC data, contact angles of solvent cast films, and elemental analysis of polymers, and full activated partial thromboplastin time results. See DOI: 10.1039/c6py00616g

thesized from acetyl protected sugar-containing diamines with hexamethylene diisocyanate (HDI). The blood compatibility of the glycopolymers were tested using anticoagulant assays, including activated partial thromboplastin time (aPTT), prothrombin time (PT) and thrombin time (TT) assays, and the best anticoagulant activities were observed with polymers with mannose or lactose functionalized polymers.

In this work we examine how changing the polymer backbone while maintaining the same sugar moieties affects blood compatibility. Lactose-containing diamines were reacted with four different diisocyanates, specifically toluene 2,4-diisocyanate (TDI), isophorone diisocyanate (IPDI), methylene bis(4-cyclohexyl isocyanate) (HMDI), and HDI, to produce a set of polyureas. The influence of degree of sulfation toward anticoagulant properties was investigated by using HMDI-based polyureas with different degrees of sulfation on the lactose. Finally, we investigated the method of action of the polymers and if they act as heparin by quantifying their ability inhibit thrombin activity in the presence or absence of antithrombin III.

Experimental

Materials and instruments

Human α-Thrombin was purchased from Haematologic Technologies (Essex Junction, VT). Antithrombin III (ATIII) was obtained from Athens Research & Technology (Athens, GA). Chromogenic substrate Chromogenix S-2366 was purchased from DiaPharma (West Chester, OH). Trizma Base was purchased through Sigma-Aldrich (St Louis, MO). All other chemicals were purchased from Sigma-Aldrich or Fisher Scientific except lactose (Matherson Coleman & Bell Co). Pre-treated standard grade, regenerated cellulose dialysis membrane with an approximate molecular weight cut off (MWCO) of 3500 g mol^{-1} was obtained from SpectrumLabs. ^{1}H and ^{13}C NMR spectroscopy was performed using a Bruker Ultrashield 400 MHz (100 MHz for ^{13}C NMR) instrument. The obtained data were processed using MestReNova software (version 6.1.0). Fourier transform infrared spectra (FT-IR) were collected on a Nicolet 6700 spectrometer and analyzed with OMNIC32 software. Mass spectrometry was performed using a Micromass Q-TOF-2TM spectrometer. Galbraith Laboratories Inc. performed elemental analyses. Molecular weight averages of polymers were determined using gel permeation chromatography (GPC) with an Agilent 1100 Series HPLC equipped with a PSS SDV Lux column (5 μm) guard column and two PSS SDV Linear XL Lux Columns (5 μm) (linear range of MW = 100 – 3×10^6 g mol^{-1}), using a filtered tetrahydrofuran (THF) containing 200 ppm 2,6-bis(1,1-dimethylethyl)-4-methylphenol (BHT) mobile phase at a flow rate of 1.0 mL min^{-1} at ambient temperature and miniDAWN TREOS light scattering (60 mW GaAs linearly polarized laser, 658 nm), Optilab rEX differential refractometer (light source = 658 nm; Wyatt Technology Corporation) and ViscoStar-II viscometer (Model NO: WV2-03, Wyatt Technology Corporation) detectors. The polymers were dissolved in eluent THF and filtered through 0.2 μm membrane filters before injection. ASTRA software v. 5.4.14 was used to determine the molecular weight averages and polydispersity. Absorbance data were collected on a Cary 50 Bio-spectrophotometer and recorded with Cary WinUV software. Blood was collected from the inferior vena cava of anesthetized C57/Bl6 mice into tenth volume sodium citrate. Platelet poor plasma was generated from this blood. All animals were maintained within federal and AAALAC guidelines. All experiments were approved by the Cincinnati Children's Research IACUC prior to being performed.

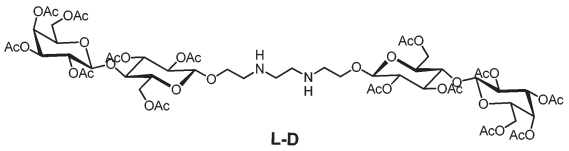


Fig. 1 Chemical structure for lactose-containing monomer L-D.

Synthesis of lactose containing diamine

The synthetic route for lactose based diamine (Fig. 1) is reported in our earlier paper.[39]

Synthesis of lactose containing polymers from L-D and TDI

Synthesis of polymer poly(TDI-pL). L-D (2.852 g, 1.0 equiv.) was added into a flame-dried flask. After evacuating and refilling with nitrogen for three times, 10 mL anhydrous DMF was injected under nitrogen protection, following by TDI (0.376 g, 1.05 equiv.). This solution was stirred at 65 °C for 40 hours before removing DMF. The residue was dissolved in a minimum amount of DCM, and precipitated out from cold ether three times. The solvent was removed under vacuum to yield 3.110 g product as a white powder in 95% yield.

Synthesis of polymer poly(TDI-L). In a flame-dried flask, poly(TDI-pL) (3.00 g) was dissolved in 40 mL anhydrous methanol and 10 mL anhydrous THF, followed by adding catalytic amount of MeONa/MeOH solution. This solution was stirred at r.t. for 2.5 h with the formation of a precipitate. The solvent was poured into a beaker and the precipitate washed with MeOH that was then combined with the original solvent. Dowex 50WX8 hydrogen form resin was added to the MeOH to adjust the pH to around 6.0. After filtration to remove the Dowex 50WX8 resin, the solvent was removed and combined with the precipitate to get product 1.78 g (94%).

Synthesis of sulfated polymer poly(TDI-sL). Polymer poly (TDI-L) (0.82 g, 1.0 equiv.) and SO_3/pyridine complex (5.44 g, 42.0 equiv.) was added into a flame-dried flask. After evacuating and refilling with nitrogen for three times, 15 mL anhydrous pyridine was injected. The reaction was heated at 90 °C for 24 h. After removing the solvent, a saturated $NaHCO_3$ solution was added to adjust the pH to 8.0 and the polymer isolated through dialysis and lyophilization to give a white foam product. Yield = 1.80 g, 91%.

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Synthesis of lactose containing polymers from L-D and other diisocyanates

The synthetic routes for L-D with other diisocyanates, IPDI, HMDI and HDI, are the same as those used for L-D and TDI. After deprotection and sulfation, the heparin-mimicking polymers were obtained. The detailed synthetic routes and full spectra are collected in ESI.†

Synthesis of sulfated glycopolymers from L-D and HMDI with different degree of sulfation

Sulfated glycopolymers from L-D and HMDI with different degrees of sulfation were synthesized by controlling the equivalents of sulfation reagent. The reaction was performed in DMF at room temperature for 24 hours. For example: the synthesis of poly(HMDI-sL)-0.57 was synthesized by mixing poly(HMDI-L) (150 mg, 1.0 equiv.) and SO_3/pyridine complex (88 mg, 2.0 equiv. for each lactose) in 6.0 mL anhydrous DMF for 24 h at room temperature. A saturated $NaHCO_3$ solution was added to adjust the pH to 8.0 and the polymer isolated through dialysis and lyophilization to give a white product.

Activated partial thromboplastin time (aPTT)

aPTTs were initiated in mouse platelet poor plasma (PPP) *via* use of Sta-PTT reagent on a STA-Hemostasis Analyzer (Diagnostica Stago). Sulfated polymers were dissolved in HEPES buffered saline (pH 7.3) and spiked into plasma prior to the initiation of the aPTT as 1/10 volume of the sample. Control specimens were spiked with HEPES buffered saline only. The final concentrations of the final sulfated polymers are as indicated in the text. Maximal observation time for the aPTT was 300 seconds.

Prothrombin time (PT)

Thromboplastin (Biopool) was reconstituted with 5 mL of water and warmed alongside wild-type mouse plasma which contained the polymer under study (prepared as for the aPTT assay) to 37 °C. The thromboplastin and plasma were mixed in a 1 : 1 ratio and agitated manually until a clot was observed. Each sample was assayed in duplicate with a maximal observation time of 60 seconds.

Thrombin time (TT)

A solution of 20 units per mL of bovine thrombin (Enzyme Research Laboratories) and 400 mM $CaCl_2$ in HEPES buffered saline was prepared and kept on ice until use. Separately, wild-type mouse plasma containing the polymer under study (prepared as for the aPTT assay) was warmed to 37 °C. The thrombin solution and wild-type mouse plasma were then mixed in a 1 : 10 ratio and agitated manually until a clot was observed to form with a maximal assay time of 60 seconds.

Thrombin inhibition experiments

Stock solution preparation. Tris·HCl buffer solution was used to prepare all solutions for the chromogenic assays, unless otherwise noted. Polymer stock solutions were prepared by dissolving lyophilized heparin or polyurea glycopolymers with buffer. Aliquots of these concentrated stock solutions were then diluted with additional buffer. The concentrations of the resulting solutions ranged from 5.0 ng mL^{-1} to 15.0 mg mL^{-1}. Thrombin solutions were prepared by diluting 1.00 μL of human α-thrombin reagent with enough buffer to give a concentration of 0.40 μM. Chromogenic substrate S-2366 was diluted to 2.0 mM with deionized water. ATIII solutions were prepared by reconstituting the lyophilized ATIII with pure water, and then further diluting with buffer to a concentration of 2 μM.

Inhibition of thrombin. Direct and ATIII-dependent inhibition of thrombin were investigated by a pseudo-first order chromogenic assay.[41–43] Polystyrene cuvettes were coated with PEG 20 000 by the method described by Latallo and Hall.[44] For direct inhibition assays, 10.00 μL of thrombin stock solution, 10.00 μL of polymer solution, and 930 μL of buffer were combined in a PEG 20 000-coated polystyrene cuvette. Solutions were incubated for 10 minutes at room temperature. 50.0 μL of S-2366 solution was then added, and the solution was stirred briefly before collecting absorbance *versus* time data at 405 nm. ATIII-dependent inhibition assays were performed in the same way as the direct inhibition assays, except that each cuvette contained 10.00 μL of ATIII stock, 10.00 μL thrombin stock, 10.00 μL polymer solution, and 920 μL of buffer. Initial rates of thrombin hydrolysis of S-2366 were obtained from the slope of the absorbance *versus* time plot. Care was taken to ensure that rates were determined at less than 10% substrate consumption, from linear portions of the plots ($R^2 > 0.999$). The activity of thrombin in the absence of polymer was determined by performing either the direct or the indirect assay under the conditions described above, except that 10.00 μL of buffer solution were used instead of 10.00 μL of polymer solution. These reference measurements were then used to calculate the fractional residual thrombin activity for each polymer concentration by dividing the activity of thrombin in the presence of polymer by the measured activity of the thrombin reference.

For direct inhibition experiments, the dose-dependence of fractional residual thrombin activities were fit to eqn (1).[42]

$$Y = Y_O + \frac{Y_M - Y_O}{1 + 10^{(\log[\text{polymer}]_O - \log IC_{50})HS}} \quad (1)$$

In this equation, Y is the fractional residual proteinase activity. Y_M and Y_O are the maximum and minimum possible values of the fractional residual proteinase activity, respectively. IC_{50} is the inhibitory concentration needed for a 50% reduction in thrombin activity, and HS is the Hill slope. For ATIII-dependent inhibition experiments, the dose-dependence of fractional residual thrombin activities were fit to eqn (2). The variables are the same as those defined for eqn (1).

$$Y = Y_O + \frac{Y_M - Y_O}{1 + \left(\frac{[\text{polymer}]_O}{IC_{50}}\right)^{HS}} \quad (2)$$

Nonlinear curve fitting was performed with Wolfram Mathematica 10.2 (Champaign, IL). The variables Y_M, Y_O, IC_{50}, and HS were allowed to float.

Results and discussion

In our previous publications,[38,39] sulfated polyurea glycopolymers with pendant glucose, mannose, lactose, or glucosamine sugars were synthesized from sugar-functionalized diamines and hexamethylene diisocyanate. It has been shown that highly sulfated glycopolymers that were designed to mimic heparin produced remarkably low complement activation.[45] We evaluated the blood compatibilities of the polyurea glycopolymers using aPTT, PT, and TT assays and the highest anticoagulant properties were seen in polymers containing sulfated mannose and lactose groups. Having screened four different sugars we now turn our attention to the influence of different diisocyanates (in other words; chemical structure and backbone flexibility) and different degrees of sulfation per sugar on the anticoagulant properties. We synthesized glycopolymers from the lactose containing diamine with TDI, IPDI, HMDI or HDI. In order to screen different degrees of sulfation we used lactose-containing polymers polymerized with HMDI. The blood compatibilities of the synthesized polyurea glycopolymers were screened using aPTT, PT, and TT assays.

Polymer synthesis and characterization

A lactose containing diamine, L-D (Fig. 1), was synthesized according to protocols we have previously reported.[28]

The polymerization of L-D with TDI was performed in anhydrous DMF at 65 °C for 40 h, as shown in Fig. 2. The number average molecular weight (M_n), weight average molecular weight (M_w) and polymer dispersity were determined to be 10 800 g mol^{-1}, 27 100 g mol^{-1}, and 2.51, respectively (Table 1).

The acetyl protecting groups in poly(TDI-pL) were removed using a catalytic amount of NaOMe in anhydrous methanol solution, producing the deprotected polymer, poly(TDI-L). The disappearance of the CH_3 signals at 1.96–2.15 ppm in the ^{1}H NMR spectrum indicated the successful deprotection of the acetyl groups, which was further corroborated by ^{13}C and DEPT-135 NMR spectroscopy. Neither primary (CH_3) nor quaternary (C=O) carbon peaks were observed in either spectrum, indicating the removal of acetyl groups from poly(TDI-pL). The deprotection reaction did not cause any cleavage of the urea groups, as evidenced by the existing peaks at 158.84 ppm for the carbon from urea functional groups and 102.87 and 102.37 ppm for the two anomeric carbon peaks. In the FTIR spectrum of poly(TDI-L) the strong ester absorption peak at 1743 cm^{-1} disappeared, and a new strong broad peak at 3311 cm^{-1} was observed due to the free hydroxyl functional groups (Fig. 3). In addition, the urea carbonyl functional group can still be observed at 1639 cm^{-1}.

The conversion of hydroxyl groups to sulfate groups was achieved using a SO_3/pyridine complex in pyridine, producing the sulfated heparin-mimicking polymer poly(TDI-sL). The observation of strong peaks at 1220, 999 and 931 cm^{-1} from the O=S=O group and the C–O–S bonds in the FTIR spectrum confirmed the successful sulfation reaction (Fig. 3). The average degree of sulfation was calculated using elemental analysis to be 7.92 sulfate groups per lactose (Table 2). While this value is higher than the theoretical maximum value (7.0 sulfates per lactose), we believe that this "over sulfation" is due to sulfonation of the electron rich phenyl ring.[46]

The same synthesis strategies were used to prepare polymers from L-D with IPDI, HMDI, or HDI. The chemical structures for the resulting polymers are shown in Fig. 4 and the molecular weight data are reported in Table 1. This allowed us to investigate the effect of structural differences in the polymer

Table 1 Gel permeation chromatography (GPC) data for the synthesized polymers[a]

Polymer	M_n [b] (g mol^{-1})	M_w [c] (g mol^{-1})	M_w/M_n
Poly(TDI-pL)	10 800	27 100	2.51
Poly(IPDI-pL)	12 300	15 200	1.24
Poly(HMDI-pL)	11 000	15 600	1.42
Poly(HDI-pL)	27 000	47 300	1.75

[a] Data were collected using THF as the mobile phase at a flow rate of 1.0 mL min^{-1} with refractive index, viscometer and light scattering detectors. [b] M_n = number average molecular weight. [c] M_w = weight average molecular weight.

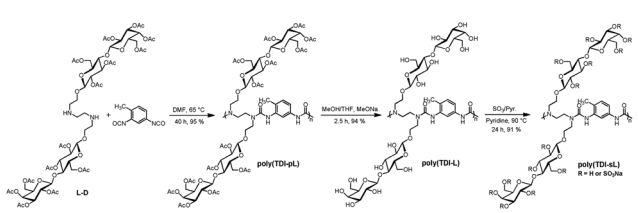


Fig. 2 Synthetic route for poly(TDI-pL), poly(TDI-L) and poly(TDI-sL).

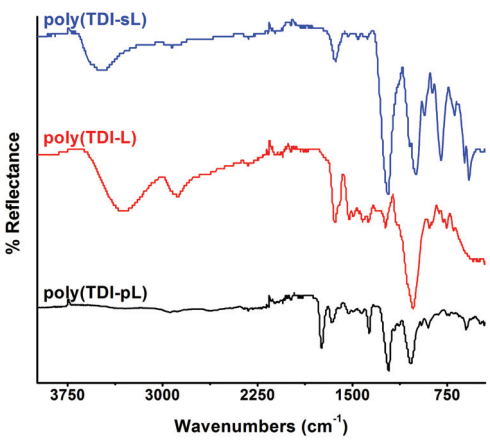


Fig. 3 FT-IR spectra for poly(TDI-pL), poly(TDI-L) and poly(TDI-sL).

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Table 2 Average degree of sulfation (DS) per disaccharide of the sulfated polymers

Polymer	DS
Poly(TDI-sL)	7.92
Poly(IPDI-sL)	6.02
Poly(HMDI-sL)	5.97
Poly(HDI-sL)	6.88

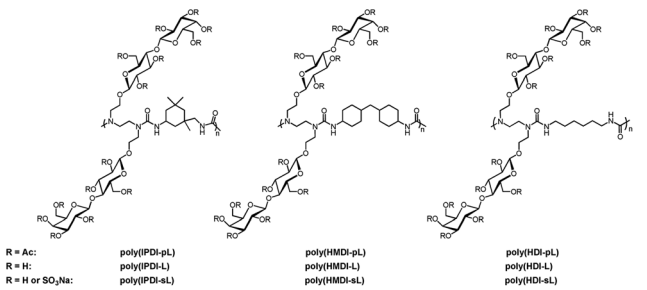


Fig. 4 Chemical structures for polymers using L-D with IPDI, HMDI and HDI monomers.

backbones on the anticoagulant properties of the polymers. The dispersity of the polymers ranged from 1.24 to 1.75. This fact, coupled with the step-growth polymerization strategy makes it difficult to explore molecular weight effects in the polymers. However, we are endeavoring to explore molecular weight effects further in intend to report these findings in future communications.

The heparin mimicking polymers, poly(IPDI-sL), poly(HMDI-sL), and poly(HDI-sL) were obtained after deprotection and sulfation of the corresponding parent polymers, and the average degrees of sulfation (DS) were calculated to be 6.02, 5.97 and 6.88 per lactose respectively. Note that in each of these cases the DS is less than seven, giving further credence to our hypothesis that the aromatic ring became sulfonated in poly(TDI-sL).

Anticoagulant assay

The aPTT assay is a standard clinical assay to characterize blood coagulation by detecting abnormalities in hemostatic function of the intrinsic and common pathways.[47] This is the most widely used assay to monitor anticoagulation with unfractionated heparin. A prolonged aPTT is consistent with anticoagulant activities of our synthesized polymers. Wild-type mouse plasma both with and without 1/10 volume HEPES buffered saline demonstrated clot formation in 28–30 s in this assay system. No prolongation of the aPTT was observed with the deprotected polymers as compared to HEPES buffered saline alone. A typical aPTT for a therapeutic concentration of heparin is between 75–115 s. In our experimental design we considered "good" clotting times to be above 100 s, as this would be consistent with therapeutic anticoagulation. The absolute values of the averaged aPTTs for the sulfated polymers are shown in Fig. 5; as the concentration of the sulfated

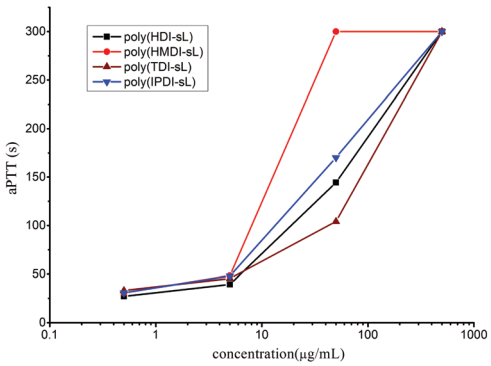


Fig. 5 aPTT results of sulfated lactose containing polyurea glycopolymers at various concentrations.

polymer increases, a corresponding increase in the prolongation of the aPTT is observed.

These results are consistent with our previous publication[39] and comparable with reported results.[48,49] For example, a sulfated lactose/acrylamide copolymer exhibited aPTTs of 100 and 200 s at concentrations of 79.9 and 184.0 μg mL^{-1} respectively.[34] A similar result was observed by Hsieh-Wilson's group,[48] who used a novel glycopolymer at a concentration of 150 μg mL^{-1} to attain an aPTT of 119.4 ± 0.5 s. The blood-compatibility of our polymers were further investigated using PT and TT assays, which corroborated the anticoagulant properties of the sulfated polymers. The results are collected in Table 3.

The deprotected polymers exhibited no anticoagulant activities in the PT assay at any concentrations, while the sulfated lactose-containing polymers did exhibit anticoagulant activity in this assay at a concentration of 50 μg mL^{-1}, except poly(TDI-sL). A HEPES buffer solution with wild-type mouse plasma was used as a negative control and afforded a PT time of 14 s, while a positive control of 0.7 IU per mL heparin solution afforded a PT of >60 s. In the TT assay the same negative control gave a TT of 21 s, and 0.1 IU per mL heparin solution used as a positive control demonstrated TTs of >75 s. The TT assay is known to be exquisitely sensitive to heparin

Table 3 Prothrombin time (PT) and thrombin time (TT) assay data in seconds with various concentrations of the synthesized polyurea glycopolymers. The data is an average of duplicate measurements and reported to the nearest 0.5 seconds

		Polymer concentration (μg mL^{-1})			
Assay	Polymer	0.5	5.0	50	500
PT	Poly(TDI-sL)	14.0	14.5	14.0	>60
	Poly(IPDI-sL)	14.5	14.0	>60	>60
	Poly(HMDI-sL)	14.0	14.0	>60	>60
	Poly(HDI-sL)	14.5	13.0	>60	>60
TT	Poly(TDI-sL)	24.5	>75	>75	>75
	Poly(IPDI-sL)	24.0	>75	>75	>75
	Poly(HMDI-sL)	26.0	>75	>75	>75
	Poly(HDI-sL)	25.0	>75	>75	>75

inhibition.[50–52] Therefore, unsurprisingly, the results from this assay (Table 3) were binary in nature, while the deprotected polymers exhibited no anticoagulant activities; there was no discrimination among the sulfated polymers as they all afforded TTs of >75 s at concentrations of 5.0 μg mL^{-1} and above.

Effect of the average degree of sulfation on anticoagulant activity

Having explored the effect of the sugar and the diisocyanate on the blood compatibility of our polymers, we turned our attention to the other main structural feature of our polymers – the degree of sulfation. All of the polymers we have prepared to date have been heavily sulfated, but do we really need such a high level of sulfation? This question is motivated in part by the pharmaceutical heparin contamination issue where the contaminant was discovered to be an oversulfated chondroitin sulfate.[53] The question is made more confounding by the varying percentage of sulfur seen in different heparin and heparin mimicking systems. For example, unfractionated heparin was reported to have sulfate percentages from 25.1% to 40.7%.[54–56] The sulfate content of low molecular weight heparins range from 30.3% to 39.0%.[56] The sulfur content of ultra-low molecular weight heparin, Arixtra, can be calculated as 14.8%. The sulfur contents of oversulfated chondroitin sulfate and oversulfated dermatan sulfate have been analyzed to be 14.4% and 12.5%, which is a degree of sulfation of 3 to 4 sulfates or 2 to 3 sulfates per saccharide respectively.[53,55,57] The Akashi group,[58] demonstrated that blood clotting time trends with the degree of sulfation of a poly(glucosyloxyethyl methacrylate) sulfate heparin-mimicking polymer. Similar results are seen elsewhere in the literature.[58–62] Therefore, correlating the degree of sulfation of our polymers with their biological response is an important task. We were able to prepare polymers with different degrees of sulfation by controlling the stoichiometric amount of the SO_3/pyridine complex we used and performing the reaction in DMF at room temperature for 24 h. The equivalents of sulfation reagent and the measured degree of sulfation by elemental analysis are summarized in Table 4.

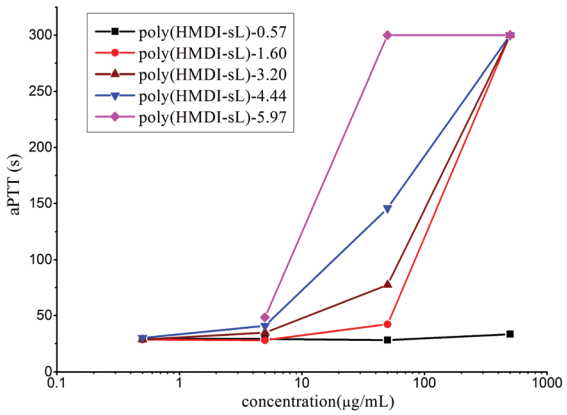


Fig. 6 aPTT of sulfated glycopolymers from L-D and HMDI with different degrees of sulfation at various concentrations.

As can be seen in Table 4, we obtained five polymers with percent sulfur ranging from approximately 3% to 15%, or an average degree of sulfation from 0.57 to 5.97 sulfates per lactose. We used the aPTT assay to test the anticoagulant effect of the average degree of sulfation of the sugars on the blood compatibility of the polymers (Fig. 6). While prolonged clotting times can be observed for all glycopolymers except for poly(HMDI-sL)-0.57 at the highest polymer concentration of 500 μg mL^{-1}, only two sulfated polymers extended clotting times to 150 s or longer at a concentration of 50 μg mL^{-1}. The order of anticoagulant activity correlates with the average degree of sulfation, *i.e.* poly(HMDI-sL)-5.97 > poly(HMDI-sL)-4.44 > poly(HMDI-sL)-3.20 > poly(HMDI-sL)-1.60 > poly(HMDI-sL)-0.57.

The PT and TT assays were performed to further probe the blood-compatibility of the sulfated glycocopolymers from L-D and HMDI with increasing degrees of sulfation, and the results are collected in Table 5. Clotting times of 15 s and 21 s respectively were observed with a HEPES buffer solution as a negative control in these assays. Heparin was again used as positive control similarly to the data described for Table 3.

Table 4 Average degree of sulfation (DS) per lactose group in poly (HMDI-L) with different equivalents of the sulfation reagent measured using elemental analysis (the numbers after the polymer name, *i.e.* 0.57, 1.60, 3.20, 4.44 and 5.97, represent the degree of sulfation of these sulfated polymers)

Polymer	SO_3·pyr.[a]	DS
Poly(HMDI-sL)-0.57	2.0	0.57
Poly(HMDI-sL)-1.60	6.0	1.60
Poly(HMDI-sL)-3.20	8.0	3.20
Poly(HMDI-sL)-4.44	10.0	4.44
Poly(HMDI-sL)-5.97	3.0[b]	5.97

[a] Equivalents of sulfation reagent per lactose unit in DMF solution used in the reaction for 24 hours at room temperature. [b] Equivalents of sulfation reagent per hydroxyl group in pyridine solution used in the reaction for 24 hours at 90 °C.

Table 5 Prothrombin time (TT) and thrombin time (TT) assay data in seconds with various concentrations of sulf(HMDI-L-D) with different degrees of sulfation. The data is an average of duplicate measurements and reported to the nearest 0.5 seconds

		Polymer concentration (μg mL^{-1})			
Assay	Polymer	0.5	5.0	50	500
PT	Poly(HMDI-sL)-0.57	15.0	15.0	14.5	15.0
	Poly(HMDI-sL)-1.60	15.0	13.5	15.5	20.0
	Poly(HMDI-sL)-3.20	14.0	15.0	19.0	>60
	Poly(HMDI-sL)-4.44	14.0	14.5	17.5	>60
	Poly(HMDI-sL)-5.97	14.0	14.0	>60	>60
TT	Poly(HMDI-sL)-0.57	22.0	24.5	24.5	37.0
	Poly(HMDI-sL)-1.60	20.0	22.0	32.0	>60
	Poly(HMDI-sL)-3.20	26.0	24.5	>60	>60
	Poly(HMDI-sL)-4.44	21.0	26.0	>60	>60
	Poly(HMDI-sL)-5.97	26.0	>75	>75	>75

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Unlike the aPTT assay where the three highest sulfated glycopolymers, poly(HMDI-sL)-3.20, poly(HMDI-sL)-4.44 and poly(HMDI-sL)-5.97, exhibited prolonged prothrombin time at the highest polymer concentration of 50 μg mL^{-1}, in the PT assay only the highest sulfated polymer, poly(HMDI-sL)-5.97, extended clotting time at 50 μg mL^{-1}.

The TT assay results are qualitatively similar to the aPTT assay. All of the sulfated polymers extended clotting times beyond the upper limit of the assay at the highest concentration except poly(HMDI-sL)-0.57. At a concentration of 50 μg mL^{-1} the three highest sulfated glycopolymers extended clotting times in the TT assay, while poly(HMDI-sL)-5.97 was also able to extend clotting times at 5 μg mL^{-1}.

Thrombin inhibition

In our previous study on the anticoagulant properties of polyureas containing sulfated carbohydrates, we found that certain polymers displayed anti-FIIa activity in both plasma spiked with ATIII, as well as plasma that had been depleted of ATIII. These findings are different to what we would expect from heparin, implying that the polymers may have a different mechanism of action than heparin. In order to investigate this we used a chromogenic assay with S-2366, a chromogenic substrate readily cleaved by thrombin, to study if our polymers interacted with thrombin in similar ways to heparin under controlled environments, *i.e.* buffer rather than biological milieu as plasma contains multiple components that can enhance or inhibit the activity of thrombin. Heparin, poly(HMDI-sL), and poly(IPDI-sL) all showed a sigmoidal dose-inhibition curve when chromogenic assays were performed in the presence of ATIII, (Fig. 7).

The general trend in IC_{50} values was heparin ≪ poly(IPDI-sL) < poly(HMDI-sL), while the trend in the maximal inhibition of thrombin activity was heparin > poly(HMDI-sL) > poly(IPDI-sL). In other words, heparin required the lowest dose to achieve 50% inhibition of thrombin activity, and was the most effective at suppressing thrombin activity in the presence of ATIII. Poly(HMDI-sL) suppressed thrombin hydrolysis of S-2366 more than poly(IPDI-sL) at the maximum value of inhibition, but required a higher concentration to reach this point. Interestingly, the two other sulfated glycopolymers with TDI and HDI did not show any inhibition of thrombin activity in the range tested. Inspecting Fig. 5 we see HMDI and IPDI based polymers gave better anticoagulant properties than HDI and TDI based polymers. We hypothesize that these results might be attributed to these polymers possessing a "moderate" flexibility, compared to the flexible polymer from HDI and the rigid polymer from TDI, therefore it is possible the HMDI and IPDI containing polymers possess enough flexibility to adopt a binding competent conformation for ATIII and thrombin with enough rigidity to maintain that conformation. Having validated the results of the glycopolymers against thrombin in the presence of ATIII, the chromogenic assays were repeated in the absence of ATIII to examine if the polymers can, unlike heparin, be direct thrombin inhibitors (Fig. 8).

Heparin, as expected, showed minimal activity against thrombin in this assay. The glycopolymer poly(TDI-sL) also showed no activity against thrombin. However, poly(HMDI-sL), poly(IPDI-sL), and poly(HDI-sL) all showed sigmoidal dose–response curves, suggesting that the polymers can directly bind with thrombin in an ATIII independent manner, albeit necessitating a larger concentration range of glycopolymer than with the presence of ATIII. The trend in maximum inhibition at saturation was poly(HMDI-sL) > poly(IPDI-sL) > poly(HDI-sL), which agrees with the order for these three polymers in the ATIII-dependent assay. The trend in IC_{50} values was poly(IPDI-sL) > poly(HMDI-sL) > poly(HDI-sL). Interestingly, poly(HMDI-sL) appears to have similar efficacy in both assays, but IC_{50} value shifts to a lower concentration in the presence of ATIII. A similar trend is shown for poly(IPDI-sL), where Y_M is similar between both assays, but the IC_{50} decreases from 4.6 μg mL^{-1} to 0.022 μg mL^{-1}. Finally, poly(HDI-sL) showed no inhibition of thrombin in the presence of ATIII for the concentration range tested, but did inhibit thrombin hydrolysis of S-2366 by approximately 20% in the absence of ATIII. Therefore, it appears that poly(IPDI-sL) and poly(HMDI-sL) may operate in a similar manner to heparin, but that these poly-

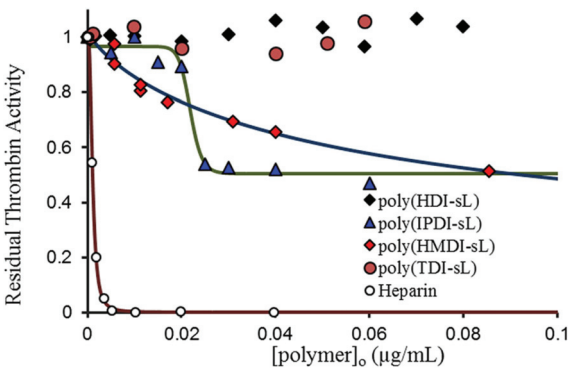


Fig. 7 Residual thrombin activity against S-2366 of heparin, poly(TDI-sL), poly(IPDI-sL), poly(HMDI-sL), and poly(HDI-sL) in the presence of ATIII. Solid lines represent the best fit to eqn (1).

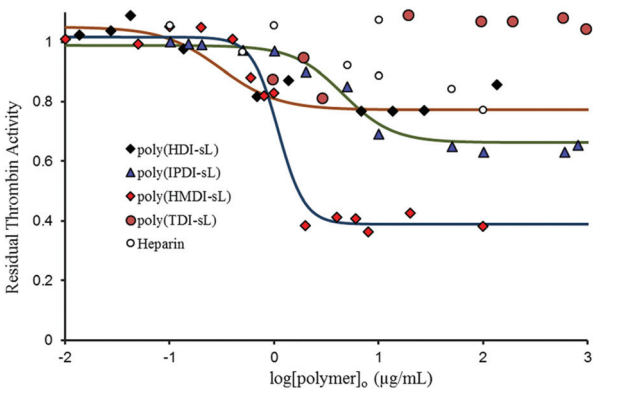


Fig. 8 Residual thrombin activity against S-2366 of heparin, poly(TDI-sL), poly(IPDI-sL), poly(HMDI-sL), and poly(HDI-sL) in the absence of ATIII. Solid lines represent the best fit to eqn (2).

mers in addition to poly(HDI-sL) can also directly inhibit thrombin (without ATIII), albeit requiring higher concentrations to exhibit an effect. While additional experiments exploring different methods of action[63–68] for each polymer are still required, this is good initial evidence that the observed aPTT trends for the polymers correlate well with the magnitude of the inhibition of thrombin in the absence of ATIII.

Conclusions

All of the highly sulfated polymers exhibited prolonged aPTT clotting times, however, the best results were observed with HMDI and IPDI comonomers. The anticoagulant activities may be attributed to the relatively flexible HMDI and IPDI containing polymers as compared with flexible HDI and rigid TDI containing polymers. These results were supported in thrombin inhibition assays, which showed that HMDI and IPDI containing copolymers afforded marked thrombin inhibition in the presence of ATIII. Interestingly, these polymers in addition to HDI-containing polymers also displayed the ability to inhibit thrombin in the absence of ATIII too, although higher polymer concentrations were required. These data taken collectively begin to allow us to rationally design heparin-inspired polymers for biomedical applications, including blood-contacting biomaterials.

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